CHITTALA UDAY KIRAN | +91-8609290186 | udaykiran2494@gmail.com

PROFESSIONAL EXPERIENCE

Onward Technologies Limited, Pune **Jul 2025 – Jan 2026**

LEAD ENGINEER

- Designed scalable ML pipelines using Random Forest, AdaBoost, XGBoost and optimization algorithms to process large datasets and significantly improve model performance for predictions.
- Built Transformer-based NLP solutions with LLMs and RAG embedding's for high-accuracy text and code generation.
- Optimized models using ensemble techniques, deploying production-ready systems with Docker for real time predictions.
- Built scalable RAG-based NLP platforms for developing domain specific AI bot assistant prototype, delivering reliable, enterprise grade AI.

Bosch Global Software Technologies, Bengaluru **Aug 2023 – Apr 2025**

SENIOR ENGINEER

- Processed unstructured load data (LKDL-12 profiles) for ML inputs, optimizing for NVH, efficiency, and deflection analysis.
- Performed feature engineering on bearing/gear datasets from MASTA simulation software and developing ML models for lifetime prediction and failure detection.
- Engineered end-to-end Machine Learning solution for e-motor bearing fault detection using ensemble model (XGBoost, AdaBoost, KNN)
- Developing RAG using Vector DB to interact with project documentation to reduce lead time and automated SOR generation for E-Axle and Electric Power Steering systems.

NBC Bearings – National Engineering Industries Limited (CK Birla Group) **Jan 2022 – Aug 2023**

EXECUTIVE MANAGER R&D – APPLICATION ENGINEER - Chennai

- Driveline Application Engineer directing customer centric Romax platform bearing solution, from requirements to system architecture, for complex Transmission, Axle and Steering systems across ICE and EV platforms.
- Lead R&D collaboration with 2W/3W/4W OEMs to define targets, drive custom design solutions, improved efficiency, NVH & durability.
- Own system level driveline models, bearing strategies and interfaces, standardizing simulation workflows and bearing across applications.
- Strategic technical partner to Daimler, Mahindra, Ashok Leyland, TAFE, Switch Mobility, Ather, TI Clean Mobility, Ola Electric, Mahindra Electric, Altigreen delivering products from concept to validation.

NRB BEARINGS PVT Ltd, Engineering Center. **Aug 2018 – Jan 2022**

ASSISTANT MANAGER R&D – APPLICATION ENGINEER – Mumbai

- Application Engineer driving Drivetrain and Steering bearing solutions from concept to customer sign-off, delivering 60+ projects approvals.
- Lead cross functional engagement with 40+ Automotive OEMs and Tier1 accounts (Tata Motors, Mahindra, John Deere, Daimler, Volvo Eicher, TAFE, Sonalika, Swaraj, Escorts Kubota), improving win rate on new RFQs through early technical influence and value selling.
- Governed a 2 year portfolio of drivetrain programs including BSVI, VAVE and light weighting, cutting design freeze cycle time and unlocking customer cost savings via optimized bearing selections.
- Steered critical failure investigations and FMEA driven simulations at customer sites, reducing repeat field issues and mentoring technical team to standardize analysis, reporting quality and stakeholder experience.

PROJECTS

AGENTIC AI - Case Study on Building and Evaluating Agentic AI Systems with LangGraph – Great Learning **Jan 2026**

- Collaborate multiple AI agents to recommend flights and hotels based on budget, quality and proximity.
- Understand agent workflows, state management and decision orchestration in action.
- Evaluated Agentic systems using goal-checking and utility-based evaluation methods.

UNIVERSITY OF TEXAS, AUSTIN (PGP in Artificial Intelligence and Machine Learning). **Jun '25 – Ongoing**

- Performed EDA with Pandas/NumPy/Seaborn/Matplotlib on food aggregator data, delivering actionable insights on restaurant/cuisine demand and customer preferences to enhance experience and boost business.
- Built a data-driven marketing model to predict personal loan uptake at AllLife Bank, uncover key customer drivers, and loan boost conversions using Scikit-Learn, Linear Regression, Decision Trees, and Clustering.
- Predicted visa approval outcomes using tuned ensemble models (Random Forest, XGBoost, Gradient Boosting), optimizing performance and identifying key factors for accurate certification or denial decisions.
- Built optimized deep learning classification models to predict wind turbine failures, reduce false negatives, and enable proactive maintenance using TensorFlow, RNN, CNN, and LSTM architectures.
- Developed a RAG-based GenAI healthcare solution using LLMs and NLP to enhance clinical decision-making, standardize care, and improve diagnostic outcomes through a functional AI Assistant.
- Built an automated helmet-detection system using CNNs (VGG-16), TensorFlow, and OpenCV to improve workplace safety compliance, accuracy, and scalable enforcement.
- Built and deployed an end-to-end SuperKart sales forecasting ML solution with robust preprocessing and tuned regression models.
- Serialized the full pipeline with joblib and exposed real-time predictions via a backend Flask REST API, containerized using Docker.
- Deployed the application on Hugging Face Spaces with an interactive frontend web interface for scalable, production-ready inference.

MASTERS THESIS PROJECT – Functionally graded polymer GRAPHENE composites – IIT KHARAGPUR **Aug '17 – May '18**

- Fabricated & designed functionally graded polymer Graphene composites & their equivalent wt. % Graphene of a bulk composite
- Improved EMI shielding effectiveness by up to 40% to 40 dB by effective stacking compared with bulk composite of same thickness
- Controlled shielding effectiveness trends by absorption & reflection phenomena measured from both ends with different wt% layers
- Validated physical samples by testing for Electrical conductivity, Scanning electron microscopy and X-Ray diffraction

SKILLS AND EXPERTISE

- **Skills**: Agentic AI, Gen AI, RAG, LLMs, NLP, Python, Deep Learning, Machine Learning, Computer Vision, CNNs, Numpy, Pandas
- **Platforms**: LangChain, LangGraph, Scikit, PyTorch, TensorFlow, HuggingFace, OpenAI, Docker, Kubernetes, AWS Cloud

COURSEWORK INFORMATION

- Artificial Intelligence
- Advanced Machine Learning
- Model Deployment.
- Machine Learning
- Natural Language Processing using Generative AI
- R Programming
- Introduction to Python
- Computer Vision
- Generative AI

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2025-26	PGP - Artificial Intelligence & Machine Learning	University of Texas, Austin	Ongoing
2018	M.TECH + B.TECH Integrated Dual Degree 5Y	IIT Kharagpur	6.42 / 10
2013	12th, Intermediate Public Examination	Board of Intermediate Education	95.80%
2011	10th, Secondary School Certificate	Board of Secondary Education	94.16%

INTERNSHIPS

NRB BEARINGS PVT Ltd, AURANGABAD **May - July, 2017**

- Initiated 20 crore project and aided in suggesting suitable furnace with layouts and technical specifications by contacting the manufacturers.
- Invited 3 furnace manufacturing company CEO's Ipsen Germany & India, Bharat Gears Limited and developed design plan.
- Proposed a single pusher continuous furnace in place of existing 3 sealed quench furnaces to ramp up production 7-8 times, thereby ruling out the need for outside side vendors and associated costs.

TVS SUNDRAM FASTENERS LIMITED, MAHINDRA WORLD CITY, CHENGALPATTU **May - July, 2016**

- Implemented Lean Manufacturing in TURBINE SHAFT 8L90 Production Line
- Identified the bottlenecks in the production line suggested techniques to reduce Cycle time to Takt time
- Improved production by 90 shafts per day in 2 CNC machines and thereby gained 120 productive minutes in a day
- Increased the process cycle efficiency using Value Stream Mapping of current state by eliminating Non-Value-Added activities and decreased the WIP time in the future Value Stream Mapping

NATIONAL THERMAL POWER CORPORATION, RAMAGUNDAM **Dec 2015**

- Studied the working of Turbines and its auxiliaries in National Super Thermal Power Station
- Analyzed fluctuation in power due to increased vibrations in turbine bearings by collecting data from data box and control unit
- Overview of Boilers, Condenser, Cooling Towers, Electrostatic Precipitators, Coal handling plant, Deaerators, Demineral plant

CERTIFICATIONS

R PROGRAMMING, COURSERA. **Aug 2016**

Enrolled and successfully completed R Programming online course offered by Prof. Roger D Peng, JOHNS HOPKINS UNIVERSITY

HEXAGON (Prev. ROMAX TECHNOLOGY). **Oct 2018**

- Successfully completed Romax Designer basic training simulation platform for engineering the next generation of electro-mechanical drive systems offered by Romax Solutions Pvt. Ltd

AWARDS AND ACHIEVEMENTS

- Awarded “Rising Champ” in NBC PROTSAHAN awards for work on ‘TML PCBU’ Gen-1 Wheel Bearing proposal delivered to customer.
- Awarded “Spotlight” for Transmission, Axle & Wheel development projects under tighter timeline with ‘Daimler’ customer.
- Awarded Silver Certificate for securing All India Rank of 185 in the Hauts Squad conducted by Technothlon, IIT Guwahati.
- Secured State Rank of 194 in the 23rd Science Talent Search Examination conducted by Dr. A S Rao Awards Council.

EXTRA CURRICULAR ACTIVITIES

- Organized the events conducted during Spring Fest 2014 as a volunteer successfully
- Member of Silver winning Meghnad Saha Hall of residence in Inter-Hall Athletics-2014
- Member of Meghnad Saha Hall of residence in Inter-Hall Football team, 2014 – 2015
- Represented in District Level Swimming competition held in jagtial, Karimnagar
- Active member of Technology Aquatics Society & NSO health and fitness, 2013 -2015, IIT Kharagpur